Recitation 10 — Answer Key

Q1)

Suppose the economy of a country is simplified to have only one good. The total annual production (nominal GDP) of this good in the current year is valued at \$2,000 billion. The average amount of money in circulation (the money supply) in the same year is \$500 billion.

1. **Calculate the velocity of money in this economy.**

   The velocity of money V is given by the quantity theory of money:

   $$MV = PY \implies V = \frac{PY}{M}$$

   where PY denotes nominal GDP and M is the money supply. For the current year:

   $$V = \frac{\$2{,}000 \text{ billion}}{\$500 \text{ billion}} = 4$$

   Each dollar is spent **four times** on average over the course of the year.

2. **What is the new velocity of money next year?**

   Applying the same formula with the updated values:

   $$V = \frac{\$2{,}200 \text{ billion}}{\$550 \text{ billion}} = 4$$

   Despite the increases in both nominal GDP and the money supply, the velocity of money remains **unchanged at 4**.

3. **Explain how a change in velocity could impact inflation.**

   From the quantity theory identity $MV = PY$, holding M and real output Y constant, an increase in V implies a higher price level P (i.e., inflation), because more transactions are occurring per dollar in circulation. Conversely, a decrease in V would put downward pressure on prices.

   In this specific example, however, V remained constant from one year to the next. In isolation, a stable velocity exerts *no* independent upward or downward pressure on inflation. The increase in nominal GDP was matched proportionally by the increase in the money supply, leaving the velocity—and thus the inflation contribution from V—unchanged.

Q2): Effective Federal Funds Rate (EFFR) Analysis

Part a)

How many recessions have there been over this time period? What happened to the EFFR during those recessions? Provide an intuitive reason.

Answer: There were **3 recessions** during the period from July 3, 2000 through April 3, 2023.

During each recession, the EFFR **fell**. This makes economic sense because:

- The Federal Reserve lowers interest rates during recessions to stimulate the economy
- Lower interest rates reduce borrowing costs for consumers and firms
- This encourages increased investment and consumption spending
- The increased spending shifts the aggregate demand (AD) curve to the right
- This policy, known as **expansionary monetary policy**, aims to increase real GDP growth
- If effective, it can help restore output to potential levels and reduce unemployment

Part b): Recent Fed Rate Changes and the AD-AS Framework

1)

As of April 3, 2023, what is the value of the EFFR? What was the value one year ago?

Answer:

- As of April 3, 2023: EFFR $\approx 4.8\%$
- One year prior (April 3, 2022): EFFR $\approx 0.3\%$

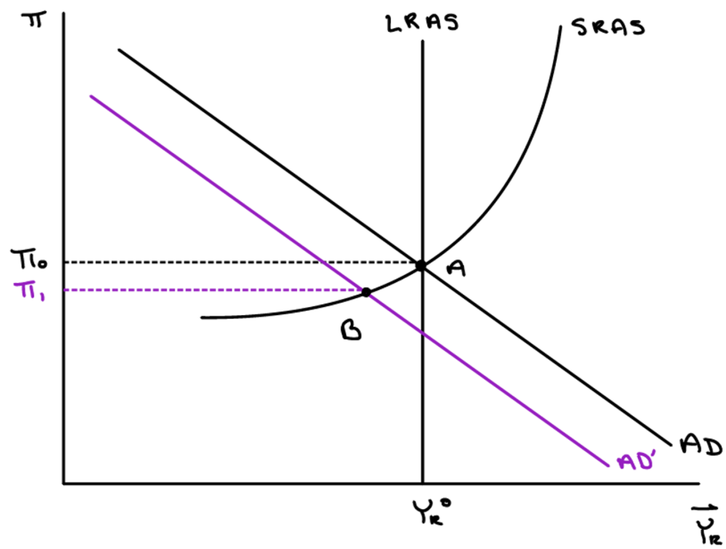
2)

Recently, the Fed has been raising interest rates because it is concerned about inflation. Using the AD-AS model, describe the logic behind the Fed's approach. Draw a graph to support your answer.

Answer:

By raising interest rates, the Fed is employing **contractionary monetary policy**. The mechanism works as follows:

1. Higher interest rates increase the cost of borrowing for households and firms
2. Investment spending falls (firms find fewer projects profitable)
3. Consumption spending falls (households find saving more attractive)
4. Aggregate demand declines; the AD curve shifts **leftward**
5. At the new equilibrium (point B), the inflation rate falls from π_0 to π_1 while real growth moderates
6. This reduces demand-pull inflation without triggering a severe recession



Part c)

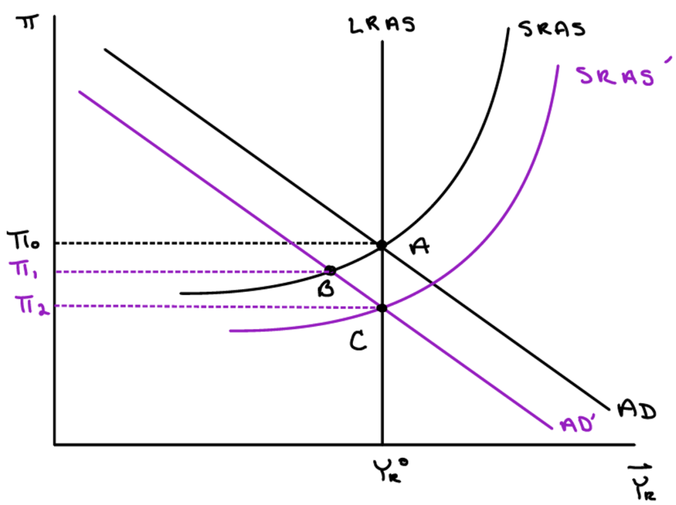
Suppose the Fed is successful in lowering the inflation rate. Explain why over time the inflation rate may fall even further. Show this change on the graph from part b.

Answer:

Once the Fed's contractionary policy begins to lower inflation, inflation *expectations* begin to decline. When households and firms expect lower inflation in the future:

1. Workers demand lower wage increases (expecting lower future prices)
2. Firms accept lower price growth expectations
3. This shifts the short-run aggregate supply (SRAS) curve **downward**

4. At the new equilibrium (point C), inflation falls further to π_2
5. Importantly, this downward SRAS shift occurs *without* further monetary tightening
6. It is a self-reinforcing mechanism: lower actual inflation anchors expectations, which further reduces inflation



Q2): Stagflation and Supply Shocks

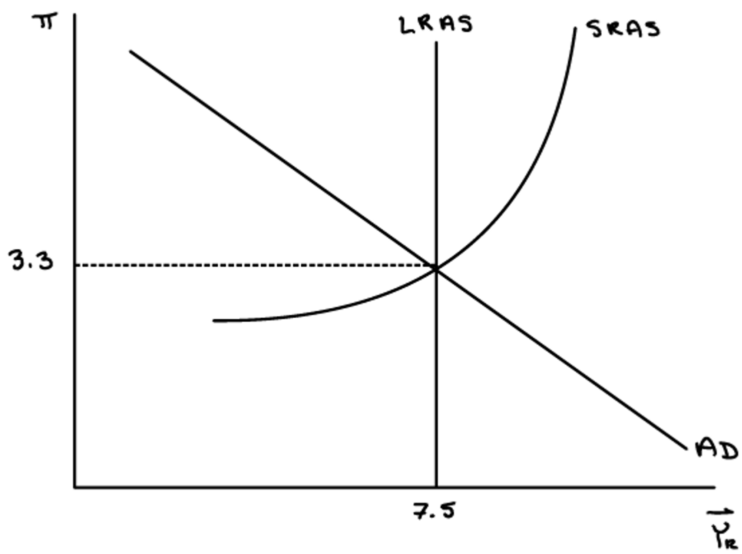
Part a)

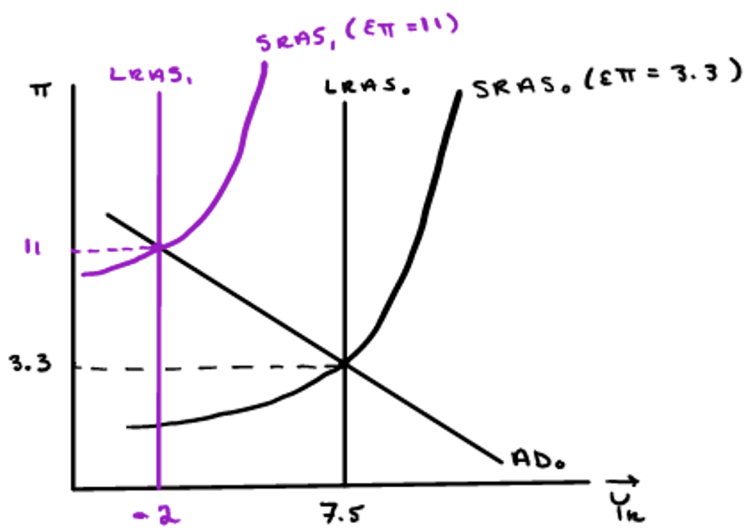
What kind of shock caused stagflation in the 1970s? Real or AD? Positive or negative? Show the impact on a graph and identify the new inflation rate (11%) and real growth rate (-2%).

Answer:
Stagflation was caused by a negative real shock (specifically, the 1970s energy crises).

- Immediate Effect (SRAS): The spike in oil prices acted as an immediate input cost shock. This shifted the Short-Run Aggregate Supply (SRAS) curve upward/leftward, as firms experienced higher production costs and cut production.
- Prolonged Impact (LRAS): Because the energy crisis was sustained, it reduced the economy's fundamental productive capacity. This caused the Long-Run Aggregate Supply (LRAS) curve to shift leftward, reflecting a new, lower potential growth rate.
- Equilibrium Shift: The intersection of the shifted SRAS and LRAS curves with the Aggregate Demand (AD) curve results in a new equilibrium point.

- Outcome: This dual shift captures the stagflation phenomenon: a simultaneous increase in the inflation rate (11%) and a decrease in the real growth rate (-2%).



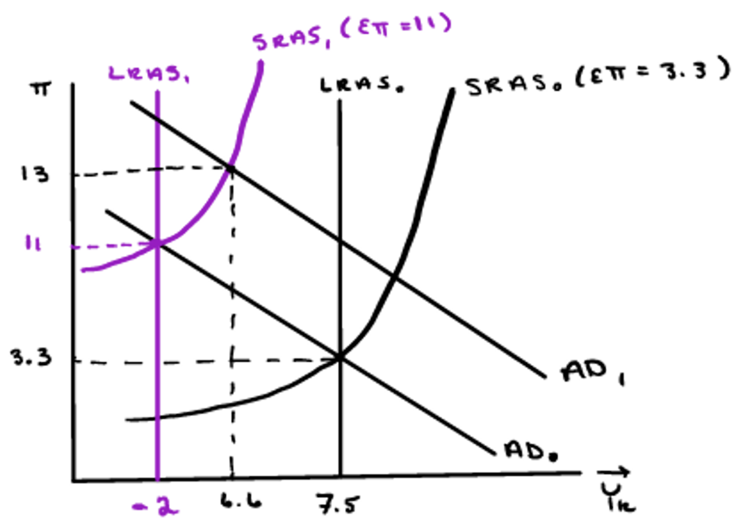


In response to these shocks, the Federal Reserve employed expansionary monetary policy to promote real growth, but this worsened inflation. Using the graph above, show how the Fed's policy increased inflation to 13% and the real growth rate to 6.6%. Which curve shifted and along which curve did the adjustment occur?

Answer:

The Federal Reserve shifted the AD curve **from** AD **to** AD_1 (rightward shift). This shift occurred **along** $\mathbf{SRAS_1}$ (the new short-run aggregate supply curve).

- Expansionary monetary policy increases the money supply
- This reduces interest rates and stimulates aggregate demand
- The movement is along the SRAS curve (not up the LRAS) because adjustment takes time
- Result: real growth increases to 6.6% but inflation worsens to 13%
- The Fed achieved short-run output gains at the cost of increased inflation
- This policy error exemplifies the inflation costs of demand-management policies when the economy faces supply constraints



Q3): Volcker's Disinflation

Part a)

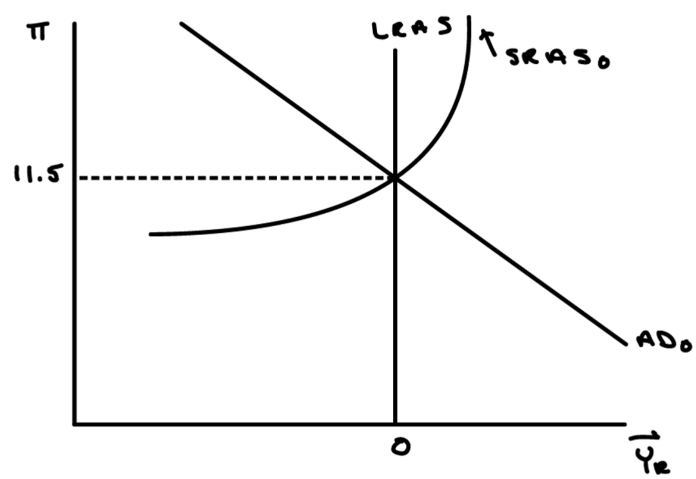
Model the U.S. economy in the late-1970s where actual inflation and expected inflation are both 11.5%, and real GDP growth is 0%.

Answer:
In the late-1970s:

- Inflation expectations (π^e) are anchored at 11.5% due to years of high inflation

- Actual inflation (π) is also 11.5% — expectations match reality
- Real GDP growth is 0% — the economy is at its potential output
- The economy is in long-run equilibrium with high inflation
- The AD and SRAS curves intersect at the LRAS

Graph:



Part b)

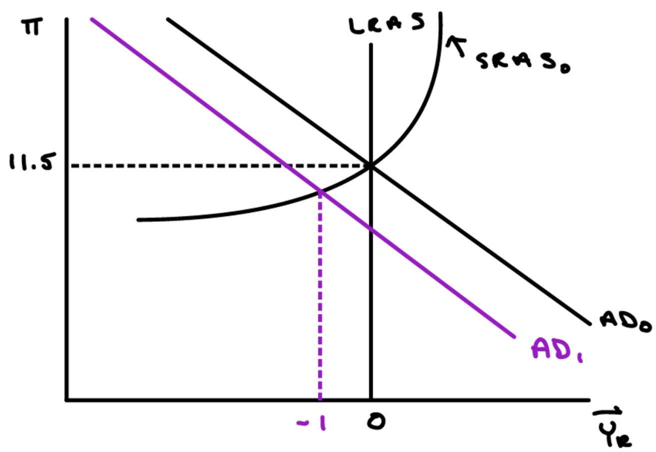
In 1979, Paul Volcker was appointed chair of the Federal Reserve Board. What impact do you think Volcker's policy had on real economic growth? Show the impact on the graph from part a.

Answer:
Volcker implemented strong **contractionary monetary policy** by sharply reducing the money supply growth rate.

- This is a left-ward shift of the AD curve from AD_0 to AD_1
- The economy moves to a point where inflation remains elevated but real growth falls sharply

- In the short run, the economy experiences a **recession**
- Real GDP growth becomes negative (the economy contracts)
- This was a deliberate, painful policy choice to break the back of high inflation

Key Insight: There is a short-run trade-off between inflation and growth. Fighting inflation requires accepting lower growth/higher unemployment in the short run.



Part c)

What impact do you think Volcker's monetary policy had on inflation expectations and actual inflation in the early-1980s? Use the graph from parts a and b to illustrate your answer.

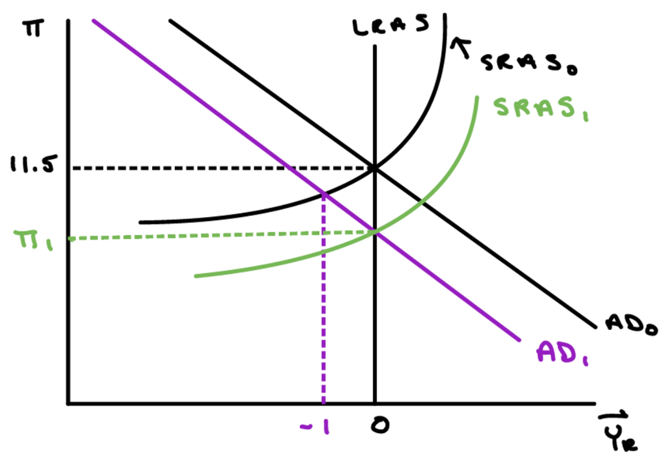
Answer:

Over time, Volcker's sustained contractionary policy succeeded in reducing inflation expectations:

1. As inflation remains lower for several periods, households and firms gradually revise expectations downward
2. Workers accept lower wage growth; firms moderate price increases
3. The SRAS curve shifts **downward** from SRAS_0 to SRAS_1
4. Actual inflation falls to π_1 (a lower level)
5. Real growth gradually recovers toward potential as inflation expectations fall

6. The economy transitions from the recession toward a new long-run equilibrium at lower inflation

Final Graph (Long-run impact):



Summary: Volcker's policy demonstrates how disinflation works in practice. In the short run, fighting inflation is costly in terms of lost growth and jobs (the recession of 1981-82). However, by maintaining the restrictive stance long enough to change inflation expectations, the economy eventually stabilizes at a lower inflation rate with restored growth. This highlights the importance of central bank credibility in anchoring expectations.